

In parametric vector form: $\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = x_3 \begin{bmatrix} 4/3 \\ 0 \\ 1 \end{bmatrix} + \begin{bmatrix} -1 \\ 2 \\ 0 \end{bmatrix}$

The equation $x = p + x_3 v$, or writing t as a general parameter,

$$x = p + tv \quad (t \text{ in } \mathbb{R}) \quad (3)$$

describes the solution set of $Ax=b$ in parametric vector form. Recall from Example 1 that the solution set of $Ax=0$ has the parametric vector eqn.

$$x = tv \quad (t \text{ in } \mathbb{R}) \quad (4) \quad [\text{with the same } v \text{ that appears in (3)}].$$

Thus the solutions of $Ax=b$ are obtained by adding the vector p to the solutions of $Ax=0$. The vector p itself is just one particular solution of $Ax=b$ [corresponding to $t=0$ in (3)]. $\square$

To describe the solution set of $Ax=b$ geometrically, we can think of vector addition as a translation. Given v and p in $\mathbb{R}^2$ or $\mathbb{R}^3$, the effect of adding p to v is to move v in a direction parallel to p to the line through p and 0 . We say that v is translated by p to $v+p$.

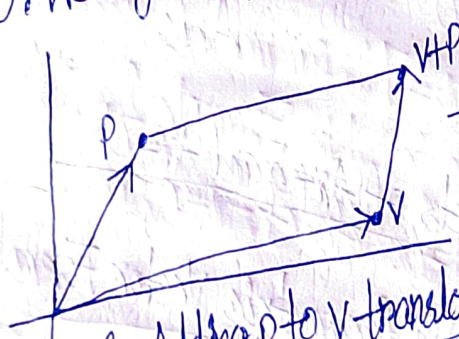


Figure 3: Adding p to v translates v to $v+p$

If each point on a line L in $\mathbb{R}^2$ or $\mathbb{R}^3$ is translated by a vector p , the result is a line parallel to L .

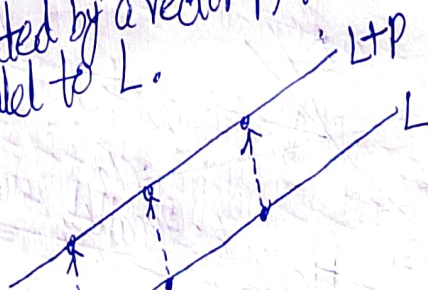


Figure 4: Translated line

Suppose L is the line through 0 and v , described by equation (4). Adding p to each point on L produces the translated line described by eqn. (3). We call (3) the equation of the line through p parallel to v . (p is on the line in eqn. (3)). Thus the solution set of $Ax=b$ is a line through p parallel to the solution set of $Ax=0$.